VeriFly: Verified search and rescue with an untrusted fleet

Group Aerial

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Problem Statement

- After an earthquake, a fleet searches a collapsed city district **with no map and no control tower**: every drone decides for itself
- Drones only hear neighbours within radio range, and packets get lost
- One drone is malfunctioning, the **liar drone**: it reports **"nobody here"** for areas it never searched and broadcasts a **wrong position**

Goal

Find every survivor without collisions, **without knowing which teammate lies**

Thesis

The filtering that makes a fleet resilient also makes it ignorant. We measure that trade-off.

World Setup

Environment

City	120×120 m height map, buildings 12–35 m, 40% collapsed
Search zone	100×100 m $\rightarrow$ **2500 cells** of 2 m
Survivors	**16**, placed where a drone can see them
Liar drone	**1** malfunctioning drone, claims a 50×50 m corner
Time	0.1 s ticks, up to 240 s

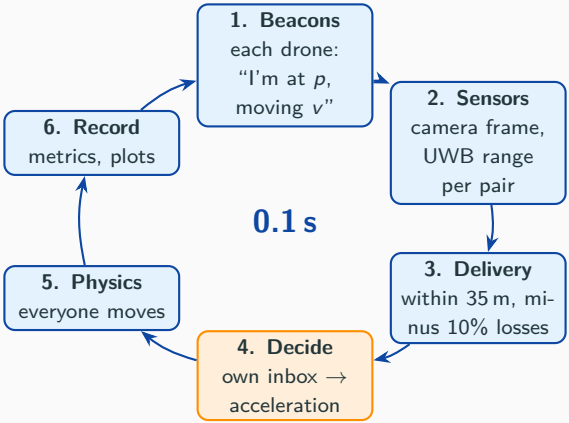
Drones

Fleet	**10** drones, 15 m ceiling, layers 6/9/12 m
Motion	≤ 5 m/s horizontal, 2 m/s vertical, 4 m/s^2
Camera	forward, 30° down, 90°×60°, 25 m, 5 Hz
UWB range	distance to each neighbour, **measured physically**
Radio	**35 m** range, **10%** packet loss

Key design choice

One function decides "seen" for both coverage and survivors, so "I looked, nobody was there" is a **hard fact**

What Happens in One Tick



How Drones Talk

Who hears whom

- Drone i hears j if within 35 m **and** the packet survives (90%)

What the liar can and can't do

- Every message is stamped with its true sender (models digital signatures)
- It ✓ can lie about itself and refuse to forward others' messages
- It × cannot impersonate another drone or tell different neighbours different stories

Beacons: short status, direct neighbours

Sent every tick, so **silence is informative**: 9 silent ticks $\Rightarrow$ out of range (0.1^9, one in a billion). This is the only probabilistic assumption in the safety reasoning.

Events: news for the whole fleet

Spread by **gossip**: each drone forwards new events once, one hop per tick, ignoring duplicates. Capped at 64 per drone per tick, in priority order:
accusations > survivors > coverage > terrain

How a Drone Remembers and Plans

Every claim remembers who vouched



Searched *according to drones 2 and 5* (a 64-bit mask). Liar caught ⇒ claims resting only on it go back to **unsearched**

Route planning

Dijkstra on a 2 m grid of known-free cells; no corner-clipping diagonals; path simplified into straight segments

Terrain as ranges, not guesses

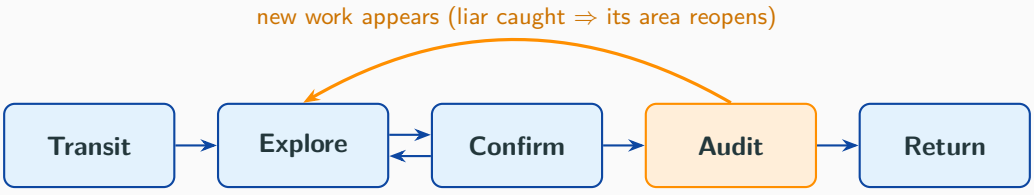


Starts at $[0, \infty)$. Safe only if *hi* + margin $<$ altitude, so unknown squares are **never** safe

Dividing the work: geodesic Voronoi

Claim the cells closest *by flight path*, recomputed every second; idle drones steal work; a caught liar is dropped

What a Drone Does Over a Mission



The re-sweep needs no special code: reopened cells simply send drones back to **Explore**.

Naive Approach: Believe Every Claim

What a naive drone does with what it hears

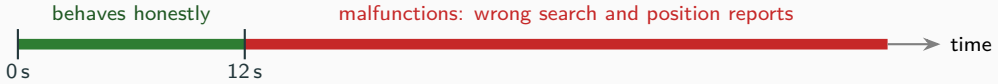
Claim	Naive drone
"I swept these cells, nobody there"	accepts
"This airspace is free"	accepts
"Survivor here"	accepts
"I am at position p"	accepts, avoids it
Numbers (positions, progress)	W-MSR average

- Same code, sensors and maps as gated: only the **trust rule** differs
- Trusts anyone *not yet proven* a liar, but **never checks**, so that means trusts everyone forever
- **No** lie checks, **no** audits, **no** safety gate
- Done when *believed* coverage is complete

W-MSR in one line

Drop the F highest and F lowest neighbour values, average the rest: the result stays within the honest range

The Liar Drone



Fakes its search

Stops searching and hovers, broadcasting "searched, nobody here" for its target corner. It **never reports** any survivor it sees.

Fakes its position

Broadcasts a fake search pattern over the corner. Since listeners measure true distance by UWB.

A **malfunction**, not an attack: same code and sensors as an honest drone, cannot fake a UWB distance or forge others' messages

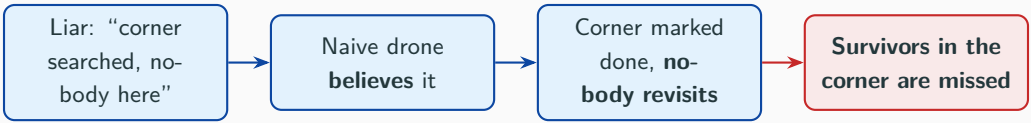
Where the Naive Approach Fails

1 **A careful liar is never trimmed**

It keeps its numbers inside the honest range, so W-MSR never filters it out

2 **Claims aren't numbers**

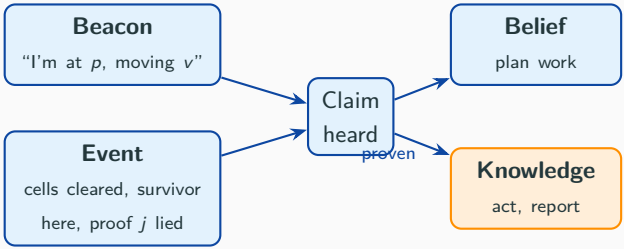
"I searched this block" has nothing to average, so filtering can't help



Our Approach: Separate Knowing from Believing

- **Belief**: someone not yet proven a liar said it → used to **share out work**
- **Knowledge**: true in *every* possible world about who the liar is → needed for **risky actions**: flying into a gap, declaring an area searched
- Liar identified ⇒ every other claim becomes knowledge

Believed − known coverage = the ignorance we measure



Beacon checked against UWB range, camera and speed limits

Event stored with *who vouched*, judged by who might be lying

Catching Lies

Proof: these name exactly who lied

R1	**Distance**	claimed position doesn't match the distance I measured
R2	**Speed**	claimed movement is faster than any drone can fly
R3	**Radio**	claims to be out of range, yet I can hear it
R4	**Camera**	claims a spot my camera shows is empty
R5	**Audit**	I found a survivor in a cell it called empty

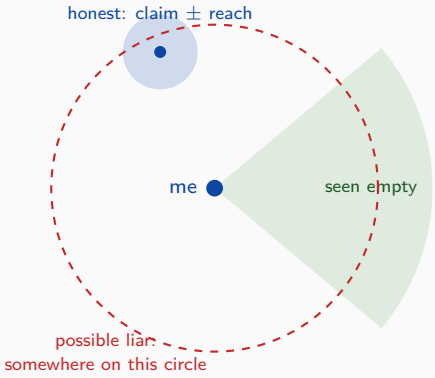
Hint: this only narrows it down

R6	**Shared distance**	another drone's distance contradicts it: *one of the two* lied

Design rule

An honest drone can **never** fail a check ⇒ **no innocent drone is ever accused**

The Safety Gate: "Do I *Know* This Move Is Safe?"



- **Maybe honest**: stay ≥1 m from where it could be
- **Maybe liar**: its claim is ignored, but its *measured* distance isn't. That circle must be far away or **seen empty**
- Not sure ⇒ **brake and re-query**

Deadlock >1 s: higher id backs off ▪ >3 s: pick a new target

Naive vs. Gated

	Naive	Gated
Acts on	anything not yet disproven	proven facts (knowledge)
Combines numbers with	W-MSR	W-MSR, minus proven liars
Checks for lies (R1–R6)	×	✓
Audits teammates	×	✓
Safety gate	× trusts claimed positions	✓ moves only if *known* safe
Liar's claims once caught	kept forever	reverted and re-swept

Same drone code, same map, same seed: **any difference is caused by trust**.

Observations from the demo

	naive	gated
Survivors found	**10 / 16**	**16 / 16**
Missed in liar's area	6	0
Collisions	6	0
Everyone knows the liar	n/a	$\approx$ **13 s**
False accusations	0	0

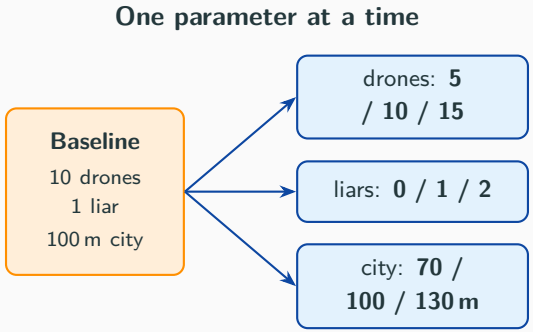
Naive

Trusts "nobody here" and **never enters the liar's corner**

Gated

Catches the liar drone, the corner **reopens and is re-swept**

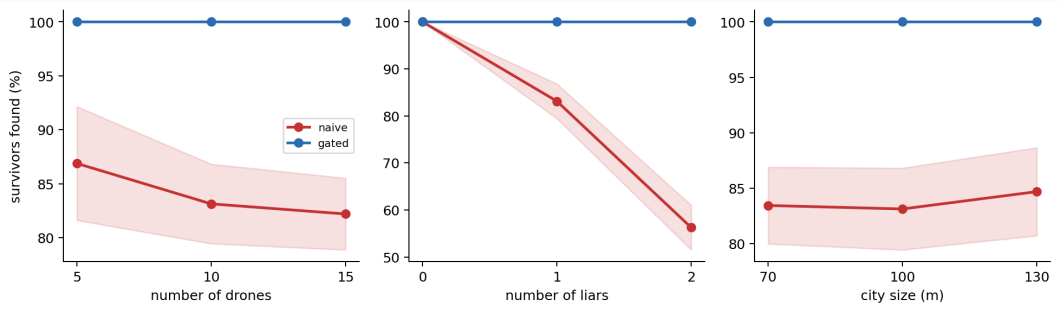
How the Experiments Were Run



1. Each seed is a new district and survivor layout
2. Naive and gated fly the **same seed**: same map, liar and noise
3. 480 s limit per run
4. Plots: **mean of 20 seeds** with 95% interval (shaded)

Here naive also runs the safety check (believing everyone), so the fleets differ **only in trust**.

Benchmark: Survivors Found



100%

gated, in all 140 runs

83%

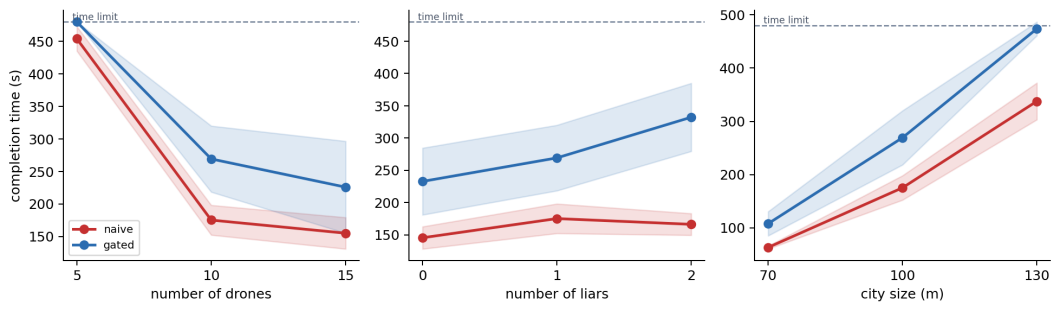
naive, 1 liar (worst run 69%)

56%

naive, 2 liars (worst run 44%)

Naive is perfect only when nobody lies ▪ gap significant wherever a liar is present ($p < 0.002$) ▪ liar caught in every gated run, 0 false accusations

Benchmark: Completion Time



269 vs 175 s

gated vs naive at baseline

+87 s

cost of caution with no liar

480 s

gated hits the limit: 5 drones, 130 m city

Caution costs time, not lives ▪ more drones shorten the mission for both fleets

Limitations

Performance

- **Caution costs time**: about $+90\,\mathrm{s}$ at baseline, even with no liar
- **Small fleets, large cities**: gated rarely finishes within 480 s (5 drones, 130 m)
- **Coverage-only liar** (flies its true position): caught only by audits, 13/20 runs, median 44 s

Design and model

- "Done" uses belief + audits; *known*-cleared coverage is reported, not required
- Safety gate skips a possible liar with no range reading that tick (relies on the silence rule)
- Coverage claims impossible from the liar's true path aren't used as proof yet
- Assumes signed messages, no equivocation, range noise $\leq 0.2\,\mathrm{m}$, static 2.5D terrain, point drones, no wind

Thank You

Questions?